

NICHOLAS BORRELLO

AI Engineer

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SUMMARY

I build production LLM systems, intelligent automation, and AI-native products. Applied AI engineer specializing in multi-provider LLM pipelines, agentic workflows, and high-throughput automation — from scraper fleets to semantic search to AI product features that ship.

EXPERIENCE

AI & Software Engineer — Baystate Pet & Garden

Jul 2024 — Present

Taunton, MA · Full-time

- Engineered a hybrid AI pipeline routing extraction between Gemini Flash and GPT-4o-mini — 100% extraction accuracy at 70% lower LLM cost than single-provider.
- Architected a distributed web-scraping fleet (Playwright + Docker) orchestrated by GitHub Actions for asynchronous vendor synchronization.
- Delivered a mission-critical inventory analytics dashboard (Next.js + Supabase) centralizing real-time sales data for daily decisions.
- Built an automated product registration engine that cut manual entry by over 80%.

NLP Data Engineer Intern — Atlas Public Policy

Nov 2023

Washington, D.C. · Internship

- Developed a spaCy-powered NLP search engine analyzing state NEVI plan documents for evidence-based policy review.
- Built a PDF-to-structured-data extraction tool that cut evidence-gathering from hours to seconds.
- Implemented fuzzy-string matching and keyword search used by the D.C. policy team for stakeholder reviews.

Process Engineering Intern — Allegro Microsystems

Jun 2023 — Aug 2023

Manchester, NH · Internship

- Designed and deployed automated CI/CD pipelines for driver testing, cutting deployment validation time by 40%.
- Introduced Infrastructure-as-Code patterns that standardized test environments across the team.

SELECTED PROJECTS

Resumancer — Agentic resume SaaS (resumancer.dev): Vercel AI SDK rewrite loop mapping job requirements to verified career evidence, reviewable staged diffs, pgvector semantic search, real-time PDF engine. Live in beta.

Baystate AI Data Pipeline — Hybrid Gemini Flash / GPT-4o-mini consolidation pipeline with validation gating; Playwright + Docker scraper fleet on GitHub Actions; Supabase/PostgreSQL product registration feeding a Next.js analytics dashboard.

ShopSite-MCP — Model Context Protocol server exposing typed, type-safe tools over a legacy ShopSite PostgreSQL schema for LLM agents — no raw SQL access.

AI PDF Search — LangChain RAG over multi-page PDFs with real-time semantic retrieval and structured answers.

Axolotl Auctions — Serverless auction platform declared entirely as AWS CDK infrastructure-as-code.

Hospital Kiosk — Frontend lead on a hospital navigation system with custom A* pathfinding and a graph-based map editor.

7Factor Staffing Tool — Enterprise staffing platform with RBAC (Auth0) and automated resource allocation; MQP capstone.

CAPABILITIES

APPLIED AI

LLM orchestration · Agentic workflows · RAG / semantic search · NLP / spaCy · MCP

APPLICATION ENGINEERING

TypeScript · Python · React · Next.js · Node.js

DATA & INFRASTRUCTURE

PostgreSQL · Supabase · Docker · Playwright · AWS CDK · Azure DevOps

EDUCATION

M.S. Computer Science — Worcester Polytechnic Institute

2020 — 2025

Worcester, MA · Specialization in Software Engineering and AI applications